

Glucose

Test Details

METHODOLOGY

Enzymatic

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Diagnose diabetes mellitus; evaluate disorders of carbohydrate metabolism including alcoholism; evaluate acidosis and ketoacidosis; evaluate dehydration, coma, hypoglycemia of insulinoma and neuroglycopenia

Custom Additional Information

According to the ADA, a fasting glucose >125 mg/dL on more than one occasion is adequate for the diagnosis of diabetes mellitus.

The ADA further classifies an intermediate group of individuals whose glucose levels do not meet criteria for diabetes yet are higher than those considered normal. These individuals were defined as having impaired fasting glucose (IFG) in plasma between 100-125 mg/dL.¹

The ADA defines three levels of hypoglycemia, with Level 1 cut-point being the most sensitive for detection.² Level 1 hypoglycemia is defined as a measurable glucose concentration <70 mg/dL but >53 mg/dL. A blood glucose concentration of 70 mg/dL has been recognized as a threshold for neuroendocrine responses to failing glucose in people with diabetes. Because many people with diabetes demonstrate impaired counterregulatory responses to hypoglycemia and/or experience hypoglycemia unawareness, a measured glucose level <70 mg/dL is considered clinically important.

Criteria for diagnosis of diabetes include²:

A1C >6.4% (utilizing NGSP certified and standardized to the DCCT assay);*
or

Fasting glucose level >125 mg/dL;* or

A two-hour plasma glucose >199 mg/dL during an OGTT;* or

Classic symptoms of hyperglycemia or hyperglycemic crisis, with random plasma glucose >199 mg/dL.

*In the absence of unequivocal hyperglycemia, criteria 1-3 should be confirmed by repeat testing.

Recent evidence revealed a diurnal variation in FPG, with mean FPG higher in the morning than in the afternoon, indicating that many cases of undiagnosed diabetes would be missed in patients seen in the afternoon. Glucose concentrations decrease *ex vivo* with time in whole blood because of glycolysis. The rate of glycolysis, reported to average 5% to 7% [~ 0.6 mmol/L (10 mg/dL)] per hour, varies with the glucose concentration, temperature, white blood cell count and other factors. Glycolysis can be attenuated by inhibition of enolase with sodium fluoride (2.5 mg fluoride/mL of blood) or, less commonly, lithium iodacetate (0.5 mg/mL of blood). These reagents can be used alone or, more commonly, with anticoagulants such as potassium oxalate, EDTA, citrate or lithium heparin. Although fluoride maintains long-term glucose stability, the rate of decline of glucose in the first hour after sample collection in tubes with and without fluoride is virtually identical. (Note that leukocytosis will increase glycolysis even in the presence of fluoride if the white cell count is very high). After four hours, the glucose concentration is stable in whole blood for 72 hours at room temperature in the presence of fluoride. In separated, nonhemolyzed, sterile serum without fluoride, the glucose concentration is stable for 14 days at 25°C and 4°C.

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection. Label specimen as serum or lithium heparin plasma.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

70–99 mg/dL

Storage Instructions

Maintain specimen at room temperature.

Patient Preparation

Blood should be drawn in the morning after an overnight fast. Per the American Diabetes Association (ADA), fasting is defined as no caloric intake for at least eight hours.¹

Causes for Rejection

Samples unspun or improperly spun (excluding oxalate/NaF); gross hemolysis; gross bacterial contamination; improper labeling

Footnotes

1. American Diabetes Association. Diagnosis and Classification of Diabetes Mellitus. *Diabetes Care*. 2014 Jan;37 Suppl 1:S81-S90. PubMed 24357215

(<http://www.ncbi.nlm.nih.gov/pubmed/24357215>)

2. American Diabetes Association. 6. Glycemic Targets: Standards of Medical Care in Diabetes-2021. *Diabetes Care*. 2021 Jan;44(Suppl 1):S73-

S84. PubMed 33298417 (<http://www.ncbi.nlm.nih.gov/pubmed/33298417>)